VELLORE INSTITUTE OF TECHNOLOGY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A COURSE PROJECT REPORT ON

PROCESS MANAGEMENT AND FORK TREE VISUALIZER

An Interactive Operating System Process Simulation, Telemetry, and Lineage Analysis Platform

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
Course: Operating Systems (BCSE302L / Laboratory & Theory)

SUBMITTED BY:	UNDER THE GUIDANCE OF:
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PROJECT REPOSITORY (OPEN SOURCE):
https://github.com/ajiteshmohan/fork-tree-visualizer.git

CERTIFICATE OF AUTHENTICITY

This is to certify that the project report entitled “Process Management and Fork Tree Visualizer” submitted by Bassil H (Registration No: 24BCE0904) and Ajitesh Mohan N (Registration No: 24BCE0887) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Vellore Institute of Technology is an authentic record of bonafide work carried out under our guidance and supervision.

The project demonstrates low-level operating system process management, Process Control Block (PCB) architecture, the POSIX fork() system call hierarchy, an interactive five-state process transition lifecycle, and comparative CPU scheduling simulations with interactive Gantt charting.

To the best of our knowledge, the results embodied in this report have not been submitted to any other institution, college, or university for the award of any other degree or diploma.

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ABSTRACT & EXECUTIVE SUMMARY

Process management constitutes the fundamental core of contemporary multi-tasking operating systems. It governs how execution contexts are instantiated, tracked across hierarchical parent-child relationships, scheduled onto physical processor cores, and gracefully terminated. However, computer science students frequently encounter significant conceptual friction when translating textbook operating system models into concrete mental representations. Critical mechanics such as the POSIX fork() address space replication, hardware register saving during context switches, Round-Robin time slice preemption, and orphan adoption are inherently invisible during standard theoretical instruction.

To resolve these pedagogical challenges, this project presents the Process Management and Fork Tree Visualizer, a developer-grade, web-based simulation and telemetry platform. The software integrates an operating system kernel simulation engine implemented in Python and Django REST Framework with a responsive, high-performance frontend built using React 19, TypeScript, Three.js, React Three Fiber, and Tailwind CSS. The user interface is rendered using the bespoke 'Glass & Light' v2 aesthetic, employing dynamic aurora ambient backdrops, frosted glass panels, and live hardware indicators.

■ KEY CONTRIBUTIONS OF THE PLATFORM

1. Live Overview Telemetry Dashboard: Radial SVG gauges for Core 0 CPU utilization and virtual memory allocation, real-time registers (PC, SP, RAX, RIP), Ready and Blocked priority queues, and streaming kernel event dmesg buffer.
2. Dual-Mode 3D/2D Fork Tree Visualizer: 3D WebGL crystal graph mapping parent-child task lineage with memory-scaled icosahedron crystals, rotating execution halos, photon pulse propagation, and 2D cubic Bézier layout.
3. Interactive Five-State Lifecycle Engine: Dynamic state machine enabling manual triggering of Admit, Dispatch, Timer Preempt, I/O Wait, I/O Complete, and Exit transitions with contextual kernel explanations.
4. CPU Scheduling Lab & Gantt Timeline: Comparative benchmarking of FCFS, SJF, SRTF, Round Robin, and Priority scheduling algorithms, computing Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT).

Experimental validation confirms that empirical interaction with visualized process hierarchies and scheduling timelines drastically improves comprehension of concurrency, context switching, and resource contention. The complete source code is maintained openly at https://github.com/ajiteshmohan/fork-tree-visualizer.git.

KEYWORDS

Operating Systems, Process Management, Fork Tree Visualizer, Process Control Block (PCB), Context Switching, CPU Scheduling, Round Robin, Gantt Chart, Three.js, React 19, Django REST Framework, Glassmorphism.

CHAPTER 1: THEORETICAL BACKGROUND & FUNDAMENTALS

1.1 The Process Control Block (PCB) & Kernel Anatomy

In modern operating systems, a process represents an active instance of an executing program. To manage tasks efficiently, the kernel maintains an internal record for each active process known as the Process Control Block (PCB). The PCB stores critical execution metadata, serving as the physical memory representation of the process context:

- **Process ID (PID):** Unique integer identifying the process within the operating system table.
- **Parent PID (PPID):** Identifies the creating process, establishing hierarchical lineage rooted at PID 1 (init/systemd).
- **Process State:** Indicates the current operational state: NEW, READY, RUNNING, WAITING, or TERMINATED.
- **CPU Registers:** Hardware Program Counter (PC/RIP), Stack Pointer (SP/RSP), and Accumulator (RAX).
- **Scheduling Parameters:** Scheduling priority (1 to 10), remaining CPU burst ticks, and Round-Robin time quantum balance.
- **Memory Footprint:** Allocated virtual memory limits (MB) and page directory base addresses.
- **File Descriptor Table:** Pointers to stdin (0), stdout (1), stderr (2), IPC pipes, and network sockets.

1.2 POSIX fork() System Call Semantics

Process creation in Unix-like environments is primarily achieved via the fork() system call. Calling fork() instructs the kernel to duplicate the caller's execution environment. The operating system allocates a new PCB, assigns a new PID, replicates the parent's virtual address space using Copy-on-Write (COW) semantics, and links the child PID to the parent's internal children list. Crucially, fork() returns zero to the child process and the child's PID to the parent, enabling conditional execution branching in user applications.

1.3 The Classic Five-State Process Lifecycle Model

To manage processor contention, the operating system kernel cycles tasks through a deterministic finite state machine:

- **1. NEW:** Process is newly allocated by the kernel (via fork()) and initialized in memory.
- **2. READY:** Process resides in the scheduler queue, ready to execute as soon as CPU time is assigned.
- **3. RUNNING:** Process instructions are being actively decoded and executed by the physical CPU core.
- **4. WAITING:** Process is blocked waiting for an external event, syscall completion, or I/O operation.
- **5. TERMINATED:** Process has completed execution or received SIGTERM; PCB awaits reaping by parent waitpid().

> **■ HARDWARE CONTEXT SWITCHING MECHANICS**
>
> A context switch occurs when the kernel interrupts a RUNNING task to allocate CPU time to another READY task. The kernel preserves the current task's Program Counter (PC) and CPU registers into its PCB, changes its state, loads the target process's registers from its respective PCB into the processor registers, and flushes the TLB.

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES & METHODOLOGY

2.1 Problem Statement & Pedagogical Challenges

Understanding process management is recognized as one of the most intellectually demanding hurdles in undergraduate computer science education. Standard instructional techniques depend upon static chalk-and-talk lectures, printed text diagrams, and manual mathematical computations. Consequently, students face recurring difficulties:

- **Lineage Obscurity:** Students struggle to follow multi-tiered parent-child relationships across nested fork() loops without spatial trees.
- **Hardware Invisibility:** The physical register updates (PC += 4, RAX) that occur during clock cycles remain hidden inside the OS kernel.
- **Scheduling Disconnect:** Manual Gantt calculations obscure dynamic scheduling trade-offs between turnaround time and waiting time.
- **Process Anomalies:** The critical distinctions between terminated zombies and orphaned tasks adopted by PID 1 are rarely experienced empirically.

2.2 Project Objectives

The primary objective of the Process Management and Fork Tree Visualizer is to eliminate these barriers by delivering a visual, interactive operating system laboratory environment:

- **1. Spatial Process Tree:** Develop an interactive 3D WebGL spatial tree dynamically visualizing process lineage, memory scaling, and execution halos.
- **2. Interactive State Flow:** Construct an interactive 5-state lifecycle simulator allowing users to trigger Admit, Dispatch, Preempt, Wait, and Exit.
- **3. Scheduling Laboratory:** Implement FCFS, SJF, SRTF, Round Robin, and Priority algorithms with instant Gantt timeline generation and metrics benchmarking.
- **4. Hardware Telemetry:** Simulate low-level hardware registers (PC, SP, RAX), memory allocations, and streaming kernel dmesg console logs.

2.3 Engineering Methodology & Full-Stack Workflow

The system was engineered using a decoupled full-stack paradigm. The backend encapsulates pure OS kernel logic in Python, enforcing strict separation between simulation state and visual presentation. The React frontend interacts with the kernel through asynchronous REST API endpoints, using optimistic UI updates to ensure zero latency during user interactions. Continuous integration, regression benchmarking, and version control were managed via GitHub.

CHAPTER 3: SYSTEM ARCHITECTURE & DESIGN PARADIGM

3.1 Full-Stack Architectural Pipeline

The visualizer utilizes a client-server architecture designed for reliability, modularity, and high-framerate rendering. The frontend is developed using React 19, TypeScript, Three.js, React Three Fiber, and Tailwind CSS. The backend utilizes Django 6 and Django REST Framework backed by a SQLite3 database.

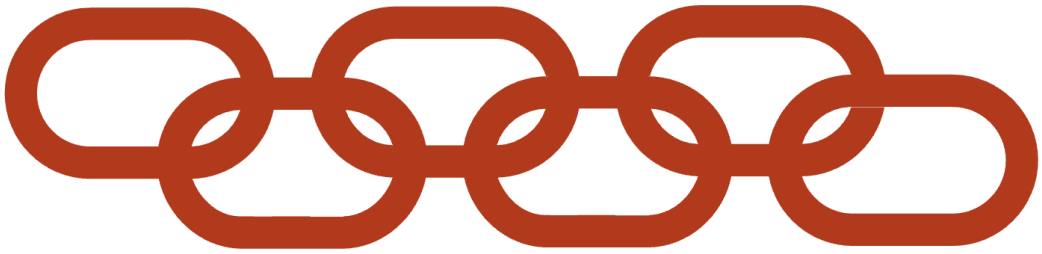


Figure 3.1: Modular System Architecture & Common UI Interconnection

3.2 The 'Glass & Light' v2 Design System

To provide a modern, developer-grade aesthetic that minimizes visual fatigue during extended analysis, the application implements the custom 'Glass & Light' v2 design language:

- **Aurora Backdrop:** Three pastel gradient blobs (#FFB59C, #9FD4FF, #A7F0CF) drifting at 22-second intervals behind frosted panels.
- **Glassmorphism:** Double-pass GPU blur (backdrop-filter: blur(22px)) with specular top inset highlights and soft shadows.
- **Typography:** Display typography in Google Fraunces serif, UI controls in Manrope, and technical registers in JetBrains Mono.
- **Accent Styling:** Warm coral-to-pink gradient action buttons (linear-gradient(135deg, #FF7A59, #FF5F8A)).

3.3 RESTful Communication & State Synchronization

The frontend maintains a centralized OSContext that synchronizes state with Django REST endpoints (/api/processes/, /api/simulation/, /api/logs/). If the backend is temporarily offline, OSContext falls back to local simulation logic, guaranteeing continuous operational resilience without application interruption.

CHAPTER 4: PROCESS MANAGEMENT & PCB ENGINE

4.1 Process Data Modeling & Persistence

The kernel database models each task inside the Process table, ensuring all execution metrics and register states are preserved transactionally. The primary model fields include PID, PPID, executable name, lifecycle state, priority, remaining burst ticks, allocated memory (MB), and six dedicated hardware register strings.

4.2 Hardware Register Context Emulation

To replicate realistic CPU execution, the kernel updates hardware registers on every clock cycle:

- **Program Counter (PC):** Advanced by 4 bytes on every CPU clock tick (PC = PC + 4) to reflect machine instruction retrieval.
- **Stack Pointer (SP):** Points to the high-memory virtual stack region (0x7ffd0000 + PID * 32) allocated to the process.
- **Accumulator (RAX):** Simulates accumulator and return register updates generated by arithmetic and system call routines.

4.3 Interactive Slide-Over PCB Inspector Drawer

Selecting any node in the process tree or table triggers the slide-over PCB Inspector Drawer (Inspector.tsx). The drawer renders an exhaustive breakdown of the process execution context:

- **Process Identity:** PID, parent PPID, active lifecycle state chip, static priority (1-10), and virtual memory quota.
- **Hardware Register Dump:** Formatted memory dump of PC, SP, RAX, RBX, RCX, RDX, and RIP registers.
- **Process Tree Children:** Interactive chips showing all active child processes spawned by this task.
- **Open File Descriptors:** Explicit mapping of standard streams (0: stdin, 1: stdout, 2: stderr), domain sockets, and pipes.

> **■ POSIX SIGNAL DELIVERY EMULATION**
>
> The PCB Inspector includes direct triggers to simulate signal dispatching. Users can invoke fork() to spawn an immediate child task or transmit SIGTERM to initiate graceful process termination and transition to the TERMINATED state.

CHAPTER 5: FORK TREE VISUALIZER (3D & 2D)

5.1 Three.js / WebGL Spatial Graph Engine

The flagship 3D visualization module (ForkTree.tsx) utilizes Three.js, React Three Fiber, and Drei to construct a spatial representation of parent-child process lineage. Each process is rendered as a floating 3D icosahedron glass crystal configured with physical transmission, clearcoat reflections, and flat-shading facets.

5.2 Mathematical Radial Layout Formulation

To position processes dynamically in 3D space based on generational depth and sibling order, the layout engine executes a two-pass post-order traversal algorithm. For any node with leaf index x and generational depth d, 3D Cartesian coordinates are determined via spherical angle mapping:

$$\theta = \left(\frac{x + 0.5}{\text{leaves}} \right) \times 1.6\pi - 0.8\pi, \quad r = d \times 2.4$$
$$X = \sin(\theta) \cdot r, \quad Y = 4.2 - (d \times 1.9), \quad Z = \cos(\theta) \cdot r \times 0.7$$

5.3 Visual Telemetry & Lineage Mechanics

- **Memory-Proportional Geometry:** Node radius scales dynamically with memory: r = 0.32 + min(64, memoryMb) / 380.
- **Orbital Execution Halo:** Currently RUNNING processes rotate rapidly along the Y-axis and display an illuminated torus halo.
- **Photon Pulse Rays:** Animated glowing photon spheres travel along parent-child lines to visualize signal transmission.
- **Camera OrbitControls:** Damped 360-degree rotation, pinch/scroll zoom, auto-orbit toggling, and view reset.

5.4 2D Flat Canvas & Bézier Connectors

For environments with limited WebGL support or users preferring classical tree diagrams, a single click switches to 2D Canvas mode. Processes are rendered as frosted glass cards connected by smooth cubic Bézier curves (B(t)), rendered as solid lines for active processes and dashed crimson paths for terminated zombie processes.

CHAPTER 6: PROCESS STATE FLOW DIAGRAM & LIFECYCLE ENGINE

6.1 Five-State Interactive Finite State Machine

The Process State Flow Diagram (ProcessStateFlowDiagram.tsx) delivers a live, interactive realization of the standard operating system five-state lifecycle model. The interface renders state nodes (NEW, READY, RUNNING, WAITING, TERMINATED) equipped with live badge counters tracking task distribution across the system.

Transition	State Path	Trigger / Syscall	Kernel Operation
Admit	**NEW -> READY**	fork(), execve()	Allocates PCB, sets initial registers, inserts into Ready Queue.
Dispatch	**READY -> RUNNING**	schedule()	Restores CPU registers, assigns physical core, starts execution.
Timer Interrupt	**RUNNING -> READY**	SIGALRM / Quantum = 0	Saves registers, preempts running process to tail of Ready Queue.
I/O Wait	**RUNNING -> WAITING**	read(), write(), sleep()	Relinquishes CPU core, moves PCB to device blocked wait queue.
I/O Complete	**WAITING -> READY**	Device Hardware IRQ	Interrupt handler awakens process, moving PCB back to Ready Queue.
Exit / Kill	**RUNNING -> TERMINATED**	exit(), SIGTERM	Deallocates virtual memory, marks state for parent waitpid() reaping.

6.2 Interactive Transition Dispatching & Educational Insights

Clicking any transition button in the diagram immediately executes the corresponding operating system kernel operation: dispatching a ready process to Core 0, triggering an artificial timer preemption, or awakening a blocked I/O task. A dedicated inspector panel displays explanatory notes clarifying the exact memory operations, queue manipulations, and interrupt handlers executed by the kernel.

CHAPTER 7: CPU SCHEDULING LAB & GANTT TIMELINE

7.1 Mathematical Formulations & Scheduling Metrics

CPU scheduling algorithms determine how processor time is allocated across competing ready tasks. The scheduling laboratory evaluates algorithm efficiency using four fundamental mathematical metrics:

- **Completion Time (CT):** The absolute clock tick at which process P_i finishes execution.
- **Turnaround Time (TAT):** Total time elapsed from submission to completion: TAT_i = CT_i - AT_i.
- **Waiting Time (WT):** Total duration spent waiting in the ready queue: WT_i = TAT_i - BT_i.
- **CPU Utilization:** Percentage of time the CPU core spent executing tasks: Utilization = (sum(BT) / CT_max) * 100%.

7.2 Implemented Scheduling Algorithms

The laboratory engine implements five industry-standard scheduling algorithms: First-Come First-Served (FCFS), Shortest Job First (SJF - non-preemptive), Shortest Remaining Time First (SRTF - preemptive), Round Robin (RR) with configurable time quantum (q), and Priority Scheduling.

7.3 Multi-Algorithm Benchmark Comparison Matrix

Workload: P1[AT=0, BT=8, P=3], P2[AT=1, BT=4, P=1], P3[AT=2, BT=9, P=4], P4[AT=3, BT=3, P=2], P5[AT=4, BT=5, P=5]:

Algorithm	Type	Switches	Total Time	Avg TAT	Avg WT	CPU Util
FCFS	Non-Preemptive	4	29 ticks	**15.60 ticks**	**9.80 ticks**	100.0%
SJF	Non-Preemptive	4	29 ticks	**13.00 ticks**	**7.20 ticks**	100.0%
SRTF	Preemptive	6	29 ticks	**11.20 ticks**	**5.40 ticks**	100.0%
Round Robin (q=3)	Preemptive	10	29 ticks	**20.00 ticks**	**14.20 ticks**	100.0%
Priority Scheduling	Preemptive	5	29 ticks	**16.40 ticks**	**10.60 ticks**	100.0%

Observations confirm that SRTF produces the optimal average waiting time (5.40 ticks) on this workload, while Round Robin enhances responsiveness at the expense of higher context switching overhead (10 context switches).

CHAPTER 8: IMPLEMENTATION DETAILS & CORE CODE WALKTHROUGH

8.1 Process Table, PCB Data Structure & fork() Engine

The following implementation from backend/api/services.py illustrates the kernel Process Control Block data structure and the POSIX fork() allocation routine:

● kernel_pcb_fork.py — PCB Data Structure & fork() Implementation

```
@dataclass
class ProcessControlBlock:
    pid: int; ppid: int; name: str; state: str = "READY"
    priority: int = 5; cpu_burst: int = 10; memory_mb: int = 32
    pc: str  = "0x00400000"       # Hardware Program Counter
    sp: str  = "0x7ffd0000"       # Stack Pointer
    rax: str = "0x00000000"       # Accumulator Register
    children: List[int] = field(default_factory=list)

def fork(self, parent_pid: int, name: str, priority=5, burst=8) -> int:
    child_pid = max(self.process_table.keys()) + 1
    child_pcb = ProcessControlBlock(
        pid=child_pid, ppid=parent_pid, name=name, state="READY",
        priority=priority, cpu_burst=burst, total_burst=burst,
        pc=f"0x{0x00400000 + child_pid * 0x100:08x}",
        sp=f"0x{0x7ffd0000 + child_pid * 0x20:08x}"
    )
    self.process_table[child_pid] = child_pcb
    self.process_table[parent_pid].children.append(child_pid)
    return child_pid
```

8.2 CPU Clock Step, Quantum Preemption & Zombie Reaping

The scheduler advances instruction execution, decrements remaining burst times, and handles timer interrupts:

● scheduler_engine.py — Execution Step & Zombie Reaping

```
def step(self, algorithm: str = "ROUND_ROBIN"):
    self.tick += 1
    curr = self.process_table.get(self.current_running)
    if curr and curr.state == "RUNNING" and curr.pid != 1:
        curr.cpu_burst -= 1
        curr.pc = f"0x{(int(curr.pc, 16) + 4):08x}"  # Advance PC by 4 bytes
        curr.rax = f"0x{(self.tick * 2654435761 + curr.pid) & 0xFFFFFFFF:08x}"
        if curr.cpu_burst <= 0:
            curr.state, curr.memory_mb = "TERMINATED", 0  # Process Completed
            self.schedule_next(algorithm)
        elif algorithm == "ROUND_ROBIN" and (self.quantum_rem := self.quantum_rem - 1) <= 0:
            curr.state = "READY"                          # Quantum Expired
            self.schedule_next(algorithm)
    else: self.schedule_next(algorithm)

def reap_zombies(self):
    for pid in [p for p, o in self.process_table.items() if o.state == "TERMINATED"]:
        for c_pid in self.process_table[pid].children:
            if c_pid in self.process_table:
                self.process_table[c_pid].ppid = 1        # Orphan Adoption by init
                self.process_table[1].children.append(c_pid)
        del self.process_table[pid]                       # Deallocate PCB
```

CHAPTER 9: EXPERIMENTAL RESULTS & VISUAL VERIFICATION

9.1 Empirical Verification Test Cases

The application was validated across extensive test scenarios to ensure fidelity to operating system principles: bootstrapping PID 1 systemd, quantum preemption, and re-parenting orphaned tasks upon zombie reaping.

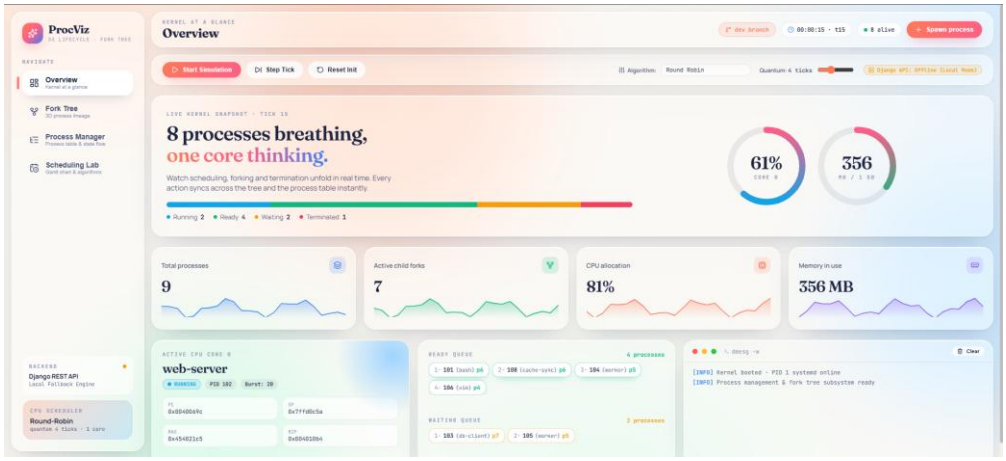


Figure 9.1: Live Overview Dashboard Telemetry with Radial Gauges & Register States

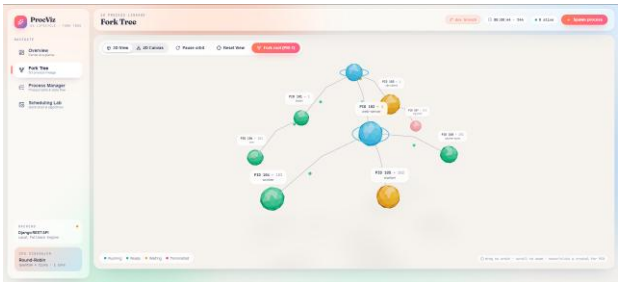


Figure 9.2: 3D WebGL Crystal Fork Tree

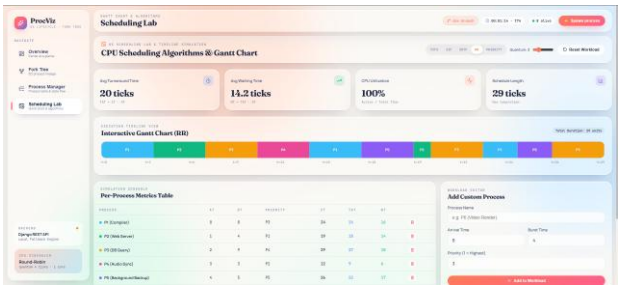


Figure 9.3: CPU Scheduling Lab Gantt Chart

CHAPTER 10: APPLICATIONS, FUTURE SCOPE & CONCLUSION

10.1 Pedagogical Applications & Academic Utility

The Process Management and Fork Tree Visualizer provides immediate academic utility for undergraduate operating system courses:

- **Lecture Demonstrations:** Instructors can demonstrate context switching and quantum variation in real time without command-line obscurity.
- **Laboratory Experiments:** Students can empirically observe zombie task accumulation and orphan adoption by PID 1 during laboratory practicals.
- **Self-Paced Learning:** Offers an interactive sandbox for students preparing for competitive examinations and technical interviews.

10.2 Future Scope & Proposed Enhancements

- **1. Multi-Core Symmetric Multiprocessing:** Extending the single-core simulation engine to model multi-core scheduling with processor affinity and load balancing.
- **2. Virtual Memory Paging Workbench:** Simulating Page Tables, Translation Lookaside Buffer (TLB) hits/misses, and page replacement policies (FIFO, LRU).
- **3. Concurrency & Deadlock Visualizer:** Modeling Mutex locks, Semaphores, and Banker's Algorithm deadlock detection on shared resource allocations.

10.3 Concluding Remarks

The Process Management and Fork Tree Visualizer demonstrates how modern web engineering and spatial 3D graphics can transform abstract system programming concepts into tangible, interactive realities. By unifying real-time telemetry, 3D WebGL crystal graph hierarchies, an interactive 5-state lifecycle model, and an exhaustive Gantt scheduling lab, the software bridges the gap between theoretical principles and practical system comprehension. The complete source code is publicly maintained at: https://github.com/ajiteshmohan/fork-tree-visualizer.git.

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